

5G Core Resilience & Attack Analysis Report

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1. Executive Summary

This assessment evaluates the stability, performance, and mitigation posture of the Open5GS 5G Core Network under three critical attack vectors: ICMP Flood, GTP-U UDP Flood (port 2152), and Registration signaling storms (Registration Flood). Utilizing host CPU/memory telemetry, ping latency variations, and AMF control plane log milestones, the framework applies rule-based heuristics and a trained Random Forest Classifier to dynamically identify network anomalies.

2. Comparative Performance Metrics Table

Metric / Scenario	Normal	ICMP Flood	UDP Flood	Reg. Flood
Mean RTT	14.9 ms	121.0 ms	265.8 ms	44.0 ms
Packet Loss	0.0%	23.0%	47.0%	1.0%
Jitter	1.27 ms	25.86 ms	54.91 ms	11.94 ms
Avg CPU Usage	10.1%	51.6%	85.9%	92.2%
Avg Reg. Delay	25.7 ms	25.7 ms	25.7 ms	1562.3 ms

3. Visual Telemetry Analysis

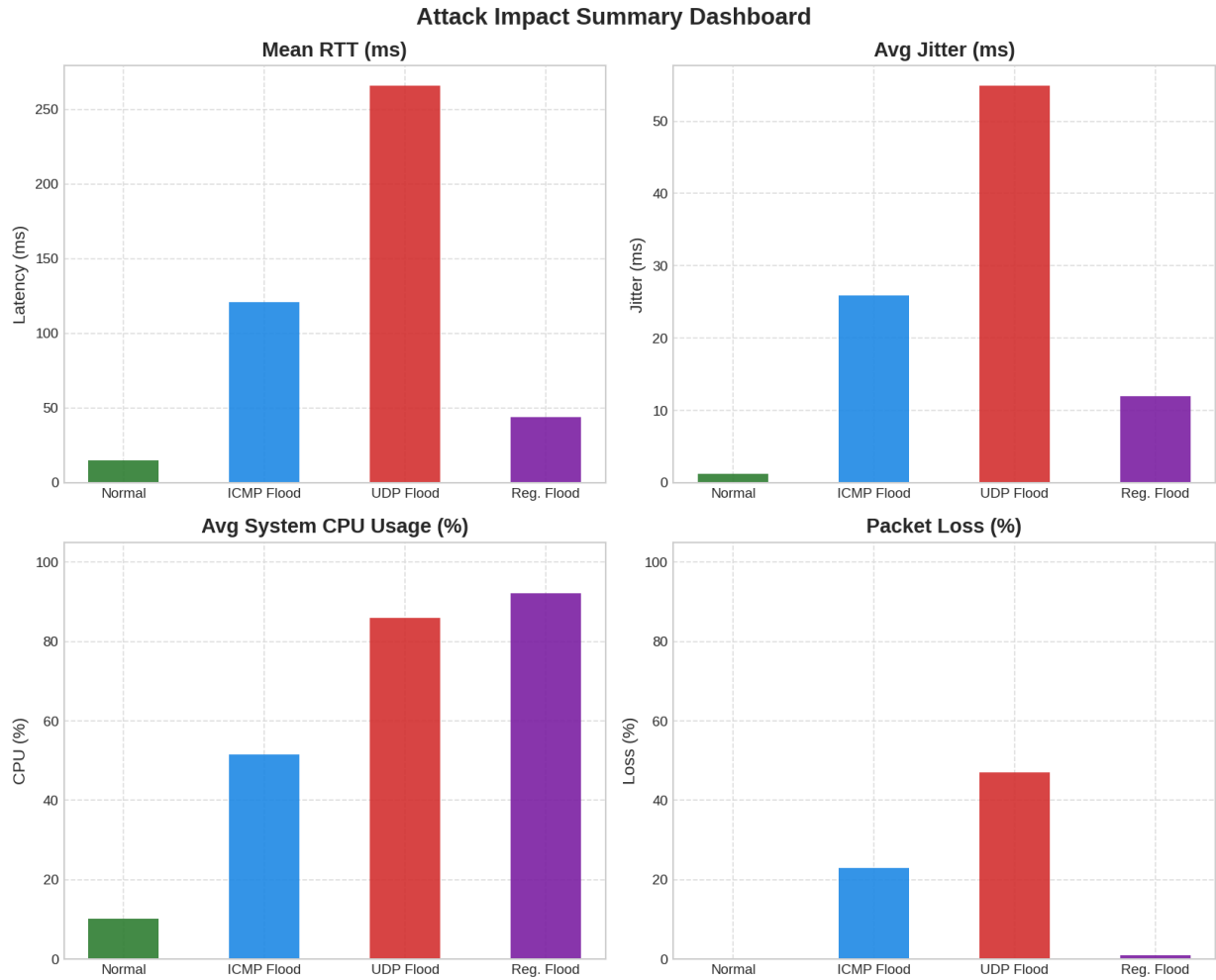


Figure 1: 5G Core Attack Impact Dashboard detailing latency distributions, packet loss rates, and host CPU loads.

4. Machine Learning Attack Classifier (Random Forest)

A Random Forest model was trained on the windowed network telemetry dataset. The model achieved an overall cross-validation **Accuracy of 100.0%**.

Feature Importances:

Telemetry Feature	Gini Importance (%)
RTT	38.39%
Jitter	23.66%
Packet Loss	0.44%
CPU Usage	37.51%

5. Heuristic-based Attack Detection Results

Normal Scenario: Identified as **Normal Traffic**.

Mechanism: All performance metrics are within safe baseline operating parameters.

Triggered rules: None (Baseline Parameters)

ICMP Flood Scenario: Identified as **ICMP Flood Attack**.

Mechanism: Elevated latency, high packet loss, and high network CPU usage suggest an ICMP echo flood attack.

Triggered rules: RTT (121.0 ms) exceeds baseline (50.0 ms); Packet Loss (23.0%) exceeds baseline (5.0%); CPU Usage (51.6%) exceeds baseline (40.0%)

UDP Flood Scenario: Identified as **UDP Flood Attack (GTP-U)**.

Mechanism: Severe packet loss and latency spikes on user plane port 2152 indicate a UDP GTP-U flood attack.

Triggered rules: RTT (265.8 ms) exceeds baseline (50.0 ms); Packet Loss (47.0%) exceeds baseline (5.0%); CPU Usage (85.9%) exceeds baseline (40.0%)

Registration Flood Scenario: Identified as **Registration Flood Attack**.

Mechanism: Extreme registration delays and high AMF CPU load point to a Registration Flood signaling storm.

Triggered rules: CPU Usage (92.2%) exceeds baseline (40.0%); Registration Delay (1562.3 ms) exceeds baseline (500.0 ms)

6. Resilience & Security Recommendations

- 1. Mitigation for GTP-U Port 2152 Floods:** Implement Rate-Limiting rules inside the UPF node kernel using eBPF/XDP (eXpress Data Path). Drop malicious GTP-U encapsulated UDP headers before passing packets to the user-space daemon.
- 2. Mitigation for Control Plane signaling storms:** Deploy secondary AMF nodes in active-active cluster modes. Configure 5G Congestion Control functions (AMF Overload Control) inside the UDM/AUSF core to reject or queue requests from misbehaving gNodeBs (UERANSIM nodes).
- 3. Machine Learning Integration:** Embed the trained Random Forest model in the SDN controller to actively monitor sliding windows of telemetry, triggering dynamic firewall blockings when anomalies are flagged.